Factorising

Algebra · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

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Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 67 minutes · Total marks: 67

KEY RULES

Common factor	$6x^2 + 9x = 3x(2x + 3)$
Quadratic	$x^2 + bx + c = (x + p)(x + q)$ where $pq = c$, $p + q = b$
Difference of two squares	$a^2 - b^2 = (a - b)(a + b)$
$ax^2 + bx + c$	Find two numbers with product ac and sum b; split the middle term
Check	Expand your answer

WORKED EXAMPLE 1

[2 marks]

Factorise $6x^2 + 9x$

Step 1	Find the highest common factor	$3x$
Step 2	Divide each term by $3x$	$2x + 3$

Answer: $3x(2x + 3)$

WORKED EXAMPLE 2

[2 marks]

Factorise $x^2 - 5x - 14$

Step 1	Find two numbers: product -14, sum -5	-7 and 2
Step 2	Write the brackets	$(x - 7)(x + 2)$

Answer: $(x - 7)(x + 2)$

WORKED EXAMPLE 3

[3 marks]

Factorise $2x^2 + 7x + 3$

Step 1	Multiply $a \times c$	$2 \times 3 = 6$; numbers 6 and 1 add to 7
Step 2	Split the middle term	$2x^2 + 6x + x + 3$
Step 3	Factorise in pairs	$2x(x + 3) + 1(x + 3)$

Answer: $(2x + 1)(x + 3)$

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1 Factorise $5x + 15$

(Total for question = 1 mark)

Practice question written for this worksheet (not from a past paper)

2 Factorise $x^2 - 49$

(Total for question = 1 mark)

Practice question written for this worksheet (not from a past paper)

3 Factorise $6x^2 + 4x$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

4 Factorise completely $12ab - 8b$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

5 Factorise $4y^2 - 25$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

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6 Factorise $16x^2 - 9y^2$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

7 Factorise $x^2 + 7x + 12$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

8 Factorise $y^2 + 6y$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

9 Factorise $x^2 - 3x - 10$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

10 Factorise $x^2 - 5x - 36$

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11 Factorise $x^2 - 12x + 32$

.....

(Total for question = 2 marks)

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12 Factorise fully $4(x-5)^2 + 3(x-5)$

.....

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13 Factorise completely $(12x - y)^2 - (4x - 3y)^2$

.....

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14 Factorise $3x^2 + 14x + 8$

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

15 Factorise $2x^2 - 5x - 12$

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

16 Factorise $5x^2 - x - 4$

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

17 Factorise fully $50g^2 - 18$

.....

(Total for question = 3 marks)

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CHALLENGE QUESTIONS

18 **14** (a) Factorise $a^2 - b^2$

.....

(1)

$N = 2^{22} - 1$

(b) Write N as the product of two integers, both of which are greater than 1000

..... × .....

(2)

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19 (a) Factorise fully $6y^2 - 5y - 4$

.....

(2)

(b) Express $\frac{2x+1}{4x} + \frac{7-5x}{3x}$ as a single fraction in its simplest form.

.....

(3)

(Total for question = 5 marks)

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20 (a) Find the value of $7n^0$ where $n > 0$

.....

(1)

b) Simplify fully $\left(2m^5p^4\right)^3$

.....

(2)

c) Factorise fully $18c^5w^2 + 30cw^3$

.....

(2)

d) Factorise $x^2 + 6x - 27$

.....

(2)

(Total for Question 9 is 7 marks)

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21 (a) Solve $5(4 - x) = 7 - 3x$

Show clear algebraic working.

$x =$ .....

(3)

(b) Factorise fully $16m^3g^3 + 24m^2g^5$

.....

(2)

(c) (i) Factorise $y^2 - 2y - 48$

.....

(2)

(ii) Hence, solve $y^2 - 2y - 48 = 0$

.....

(1)

(Total for question = 8 marks)

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22 (a) Factorise $6y^2 - y - 5$

.....

(2)

(b) Make f the subject of $w = \frac{2f + 3}{8 - f}$

.....

(3)

(c) Express $4x^2 - 8x + 7$ in the form $a(x + b)^2 + c$ where a, b and c are integers.

.....

(3)

(Total for question = 8 marks)

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TOTAL FOR PAPER IS 67 MARKS